

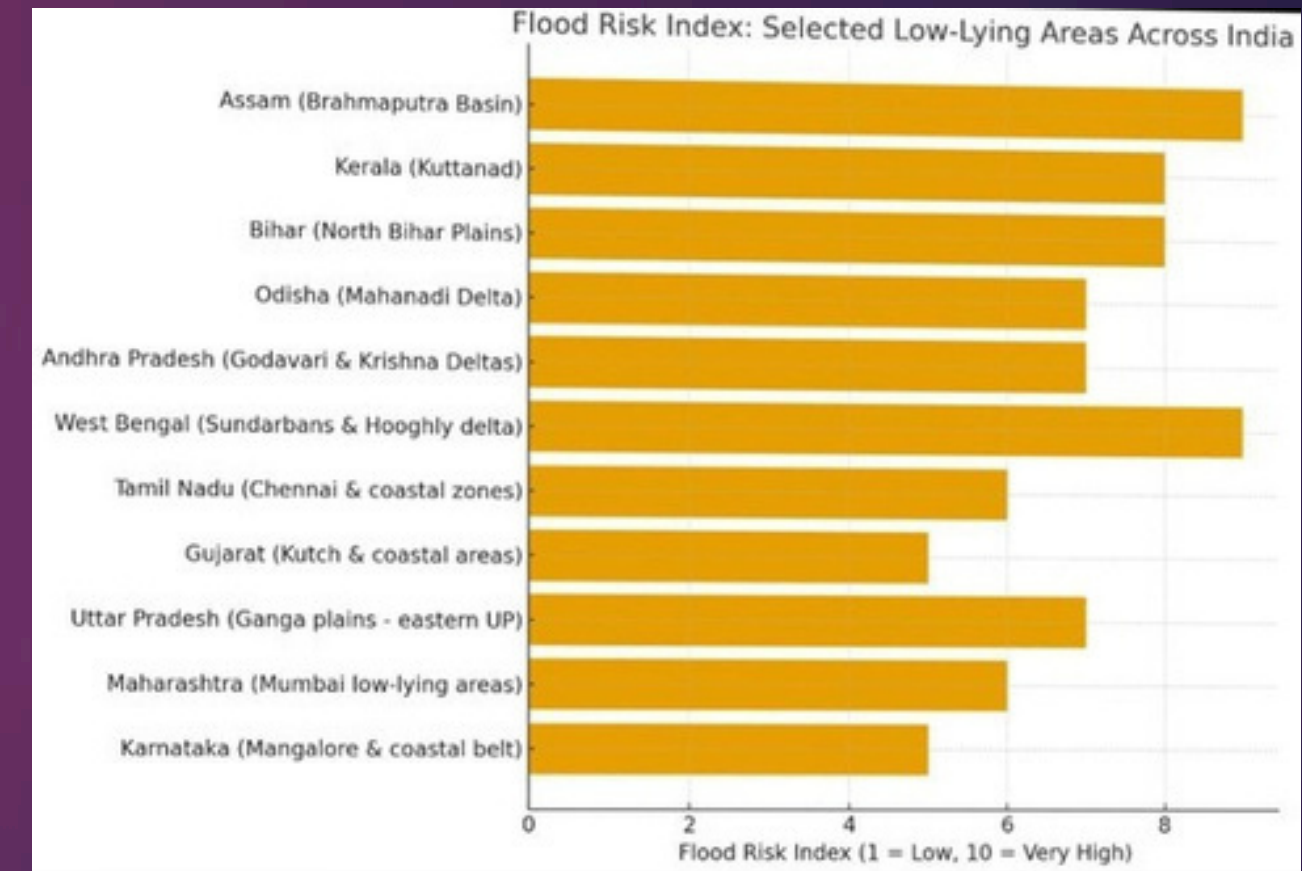
SMART EVAC

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PROBLEM :

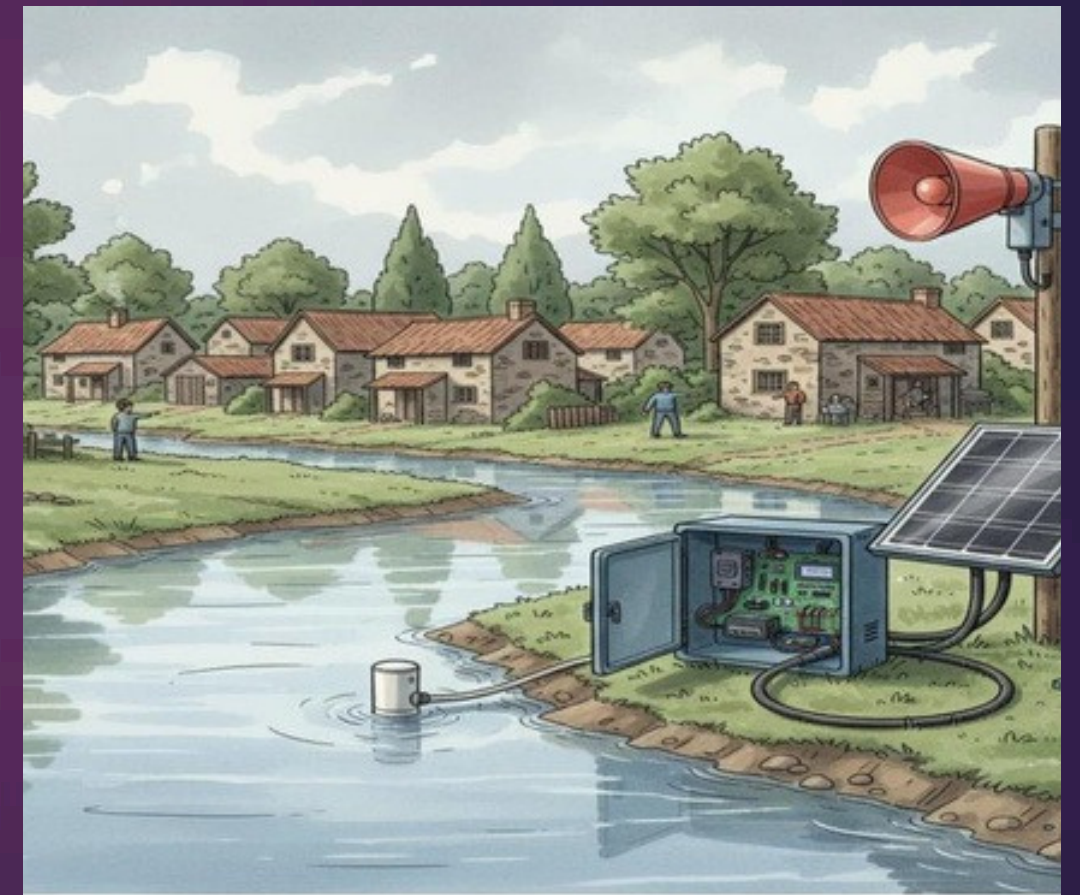
Low-lying areas are highly susceptible to flooding, causing:

- 1 property damage and crop loss.
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- 2 Displacement of families and livestock. Disruption of transportation and daily life. Traditional flood warnings are often delayed or too general, leaving little time to react.
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- 3
-
- 4 Communities lack a localized, proactive system to safely relocate people and assets before floodwaters arrive.
-
- 5
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SOLUTION :

- A community-driven IOT based flood monitoring system designed to detect rising water levels in low-lying areas and trigger early evacuation alerts. Features include:
- Water level & rainfall monitoring using IOT sensors.
- Real-time alerts to residents via mobile app, SMS, or WhatsApp.
- Safe evacuation plans activated automatically when flood thresholds are reached.
- Crowdsourced reporting for faster verification and updates.
- Historical data collection for future urban planning and flood mitigation



TECHNOLOGY USED:

Hardware

- Water sensors (ultrasonic, pressure based) → to measure how high the water is.
- Rainfall sensors → to check how much it is raining. Small computers (Arduino, ESP32, Raspberry Pi) → to collect and send the sensor data.
- Solar panels with batteries → so the system keeps working even during power cuts.
- Internet connection (GSM, LORA, or Wi-Fi) → to send information to the server.

Software

- Cloud → stores data
- Dashboard → shows water & shelters
- Alerts
- GPS → SMS, WhatsApp, app
- → shows nearest safe place
- AI → predicts flood risk



Components

Arduino

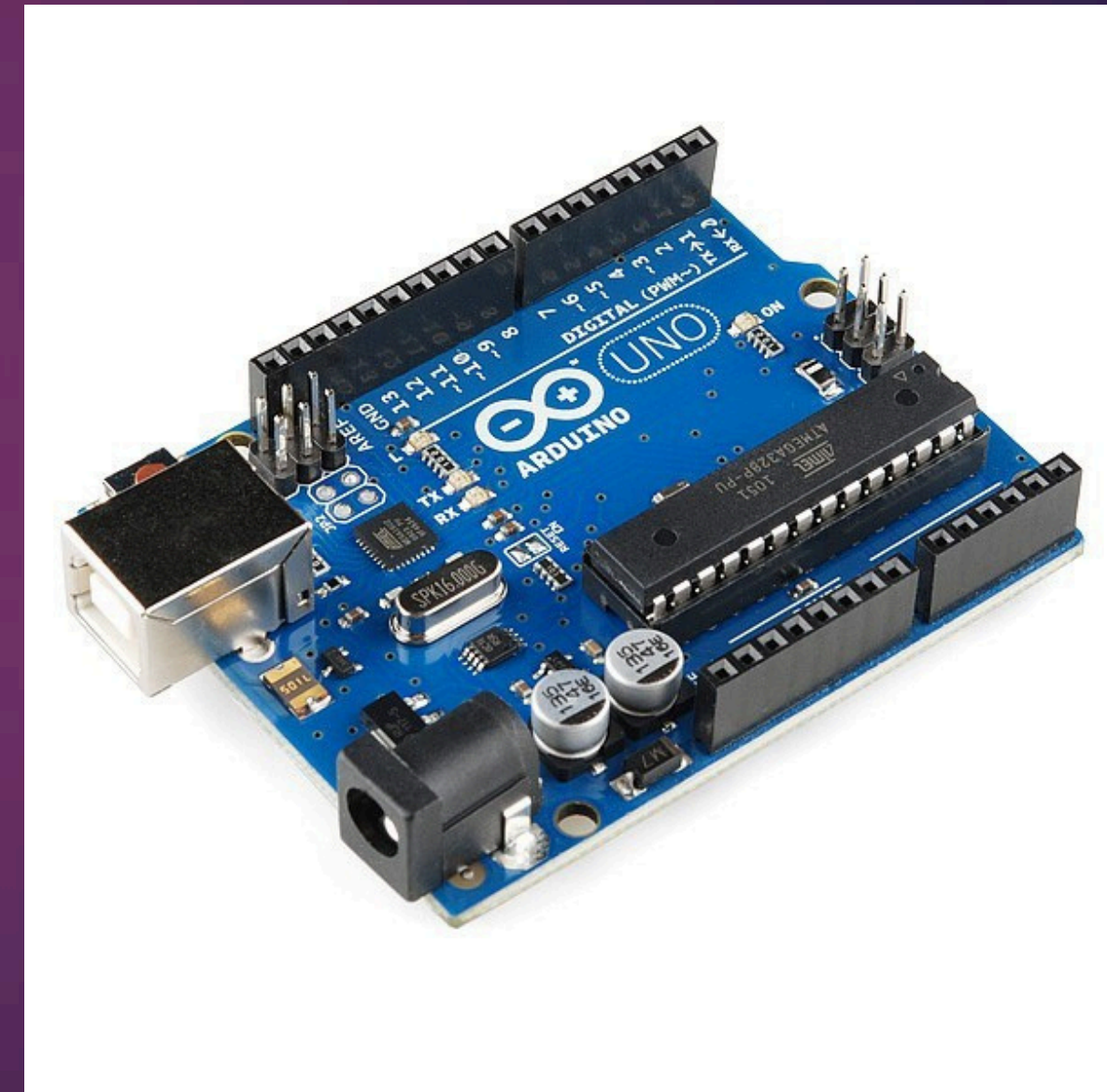
Arduino is an open-source microcontroller board used for building electronic and IoT projects. It reads inputs from sensors and controls outputs like LEDs, motors, and buzzers.

Key Features

- Based on ATmega328P (Arduino UNO)
- 14 digital pins, 6 analog pins, 6 PWM pins
- Operates at 5V, programmable via USB
- Supports UART, I2C, and SPI communication

How It Works

- Input: Reads data from sensors
- Process: Executes the uploaded program
- Output: Controls devices based on logic.



○ ESP32

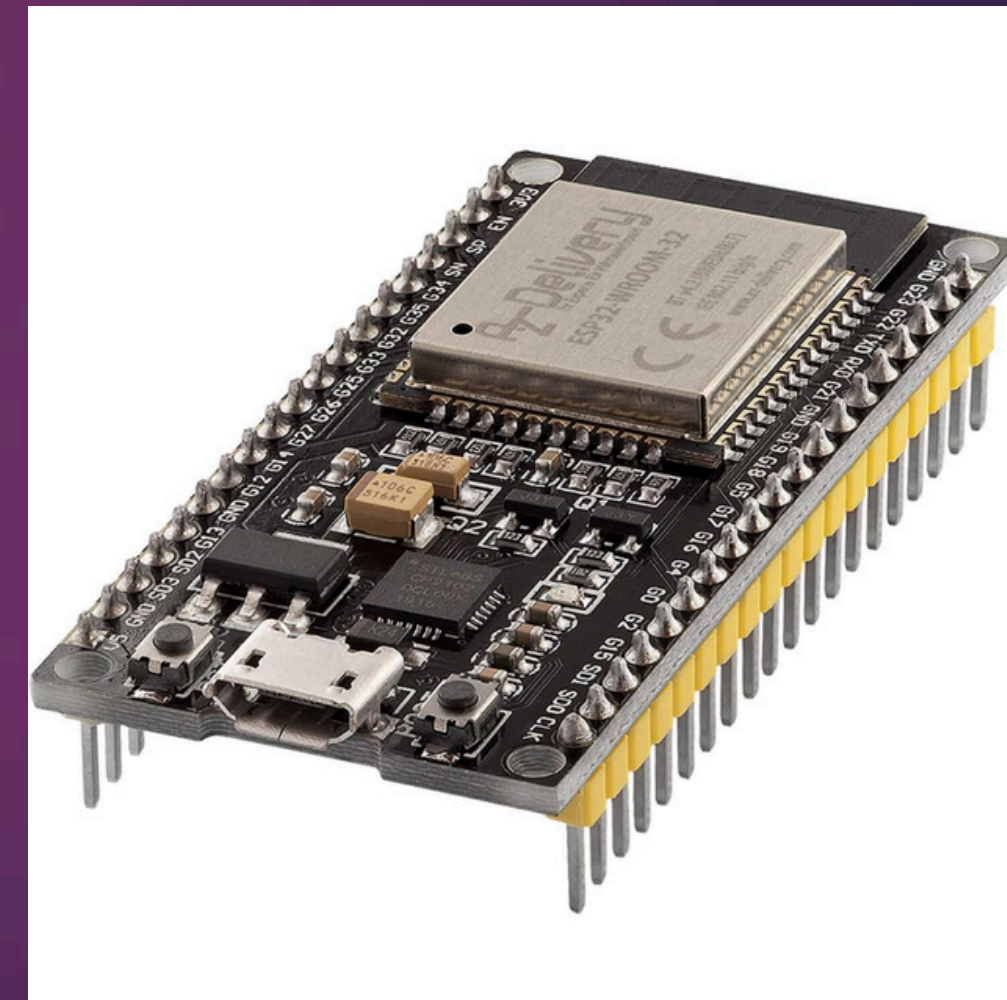
- ESP32 is a powerful Wi-Fi and Bluetooth-enabled microcontroller used for IoT and smart automation projects.
- It offers high performance, low power consumption, and multiple connectivity options.

○ Key Features

- Dual-core 32-bit processor
- Built-in Wi-Fi (2.4 GHz) and Bluetooth/BLE
- Multiple GPIO, ADC, PWM, UART, SPI, I2C pins
- Operates at 3.3V
- On-board timers, touch sensors, and DAC

○ How It Works

- Reads data from sensors
- Processes using uploaded code
- Sends/receives data using Wi-Fi/Bluetooth
- Controls connected devices (LEDs, motors, relays)



○ Ultrasonic Sensor

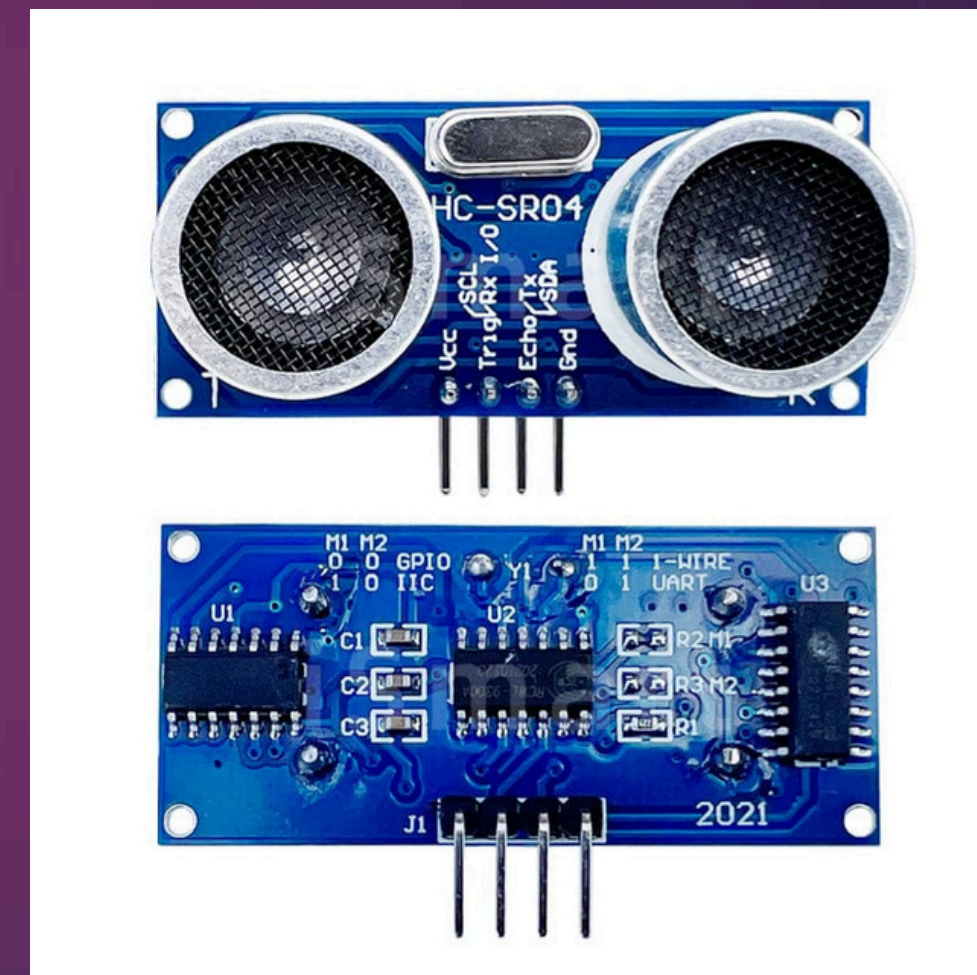
- An ultrasonic sensor measures distance using high-frequency sound waves.
- It sends an ultrasonic pulse, waits for the echo, and calculates distance based on the time taken to return.

○ Key Features

- Model: HC-SR04
- Range: 2 cm – 400 cm
- Frequency: 40 kHz
- Pins: VCC, GND, TRIG, ECHO

○ How It Works

- Arduino/ESP32 sends a 10 μ s trigger pulse
- Sensor emits ultrasonic sound
- Sound hits an object and returns to the sensor
- ECHO pin gives the time taken
- Distance = (Time $\times$ Speed of Sound) / 2



- **Resistor & Buzzer**

- Resistor & Buzzer – Overview (Single Slide Content)

- Resistor

- A resistor is an electronic component that limits or controls the flow of electric current in a circuit.

- It protects components like LEDs, buzzers, and sensors from excessive current.

- Key Points

- Measured in Ohms (Ω)

- Follows Ohm's Law ($V = IR$)

- Prevents damage due to high current

- Color bands indicate resistance value

- **Buzzer**

- A buzzer is a sound-producing device used for alerts, alarms, and notifications in electronic projects.

- Key Points

- Types: Piezo buzzer (passive), Active buzzer (built-in oscillator)

- Works using vibration of a diaphragm

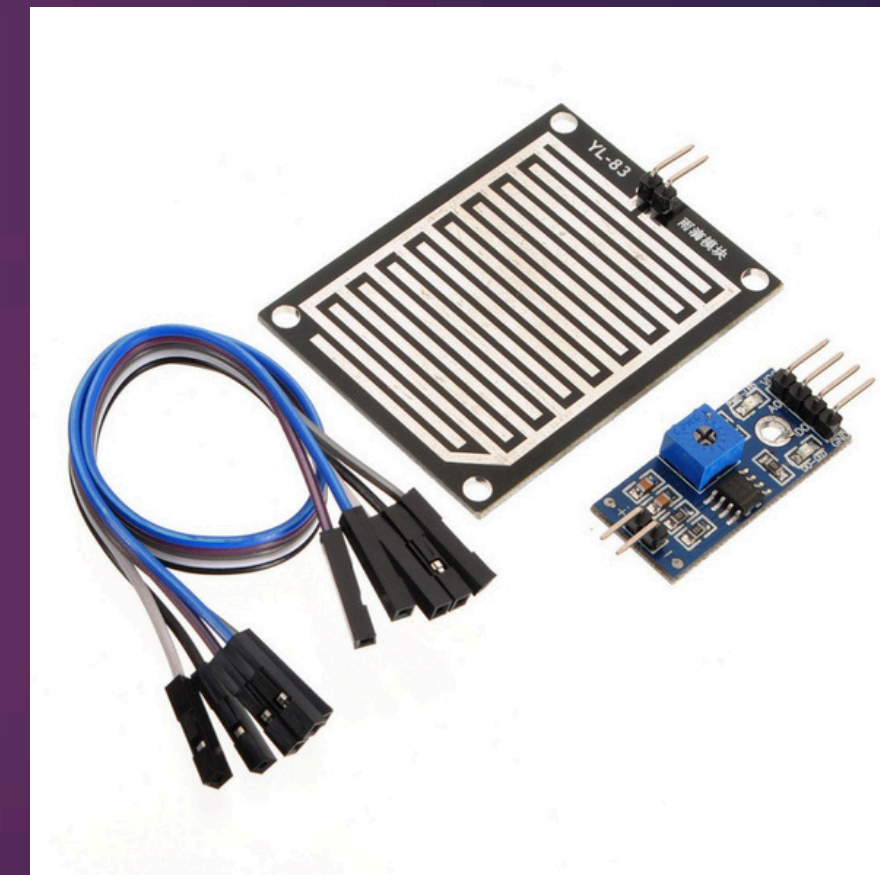
- Operates typically at 5V

- Controlled by Arduino/ESP32 using a digital pin



RAIN SENSOR

- A RAIN SENSOR DETECTS THE PRESENCE AND AMOUNT OF RAIN BY MEASURING CHANGES IN CONDUCTIVITY ON ITS SENSING PLATE.
- IT IS COMMONLY USED IN SMART WEATHER, IRRIGATION, AND AUTOMATION SYSTEMS.
- KEY FEATURES
 - CONSISTS OF A RAIN SENSING PLATE AND CONTROL MODULE (LM393)
 - OUTPUTS:
 - ANALOG (AO): RAIN INTENSITY
 - DIGITAL (DO): RAIN DETECTED / NOT DETECTED
 - OPERATING VOLTAGE: 3.3V–5V
- HOW IT WORKS
 - WATER DROPLETS FALL ON THE SENSOR PLATE
 - CONDUCTIVITY INCREASES AND RESISTANCE DECREASES
 - THE MODULE CONVERTS THIS INTO ANALOG AND DIGITAL SIGNALS
 - MICROCONTROLLER READS THE SIGNAL TO DETECT RAIN LEVEL



LED

Light-emitting diode (indicator light)

Two legs: + (anode) and – (cathode)

Works with a resistor to avoid damage

Shows status or alerts in circuits



Software components

1. Cloud Server

Stores all the data coming from sensors.

2. Dashboard (Web or Mobile App)

Shows live water level and rainfall information.

3. Alert System

Sends SMS, WhatsApp or app notifications to people if water level is dangerous.

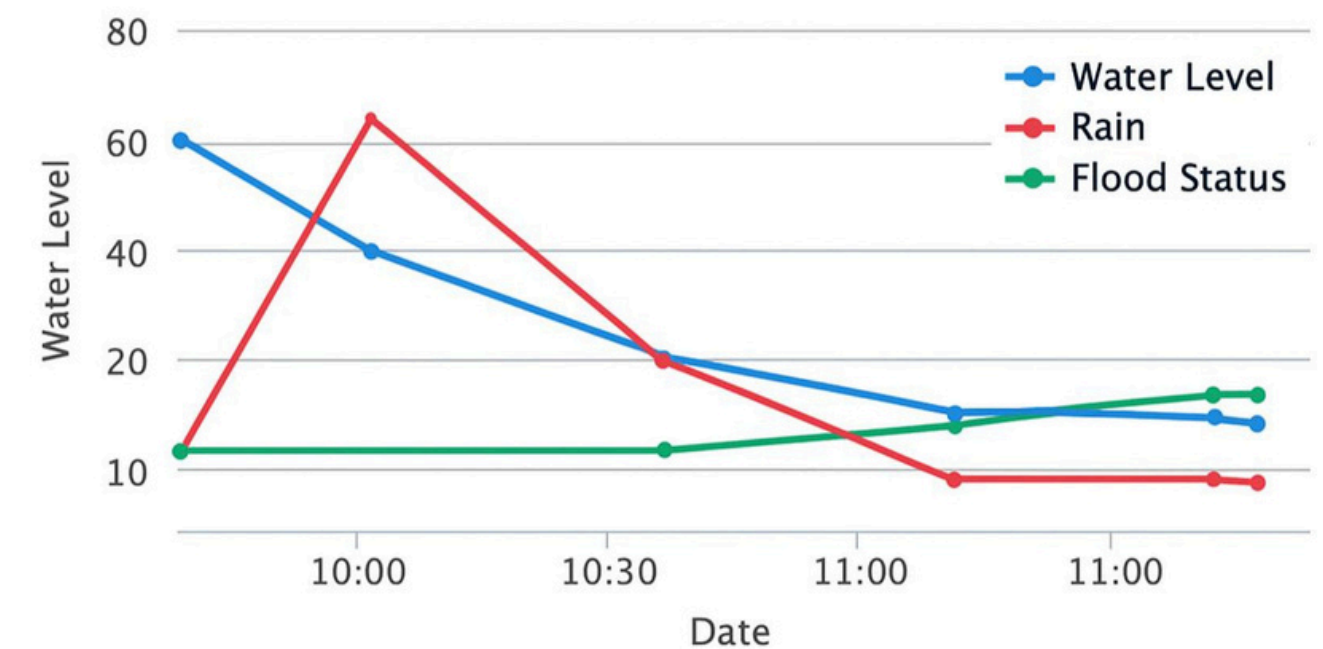
4. GPS Integration

Helps people locate the nearest safe shelter.

5. AI Prediction

AI analyzes previous and live data to predict possible flood risks.

Field 2 Chart



Connections

1. Arduino UNO Pin Connections

◆ Ultrasonic Sensor (HC-SR04)

Sensor Pin Connect to Arduino

VCC 5V
GND GND
TRIG Pin 9
ECHO Pin 10

◆ Rain Sensor (Analog Type)

Sensor Pin Connect to Arduino

VCC 5V
GND GND
AO A0

◆ LED

LED Pin Connect to Arduino

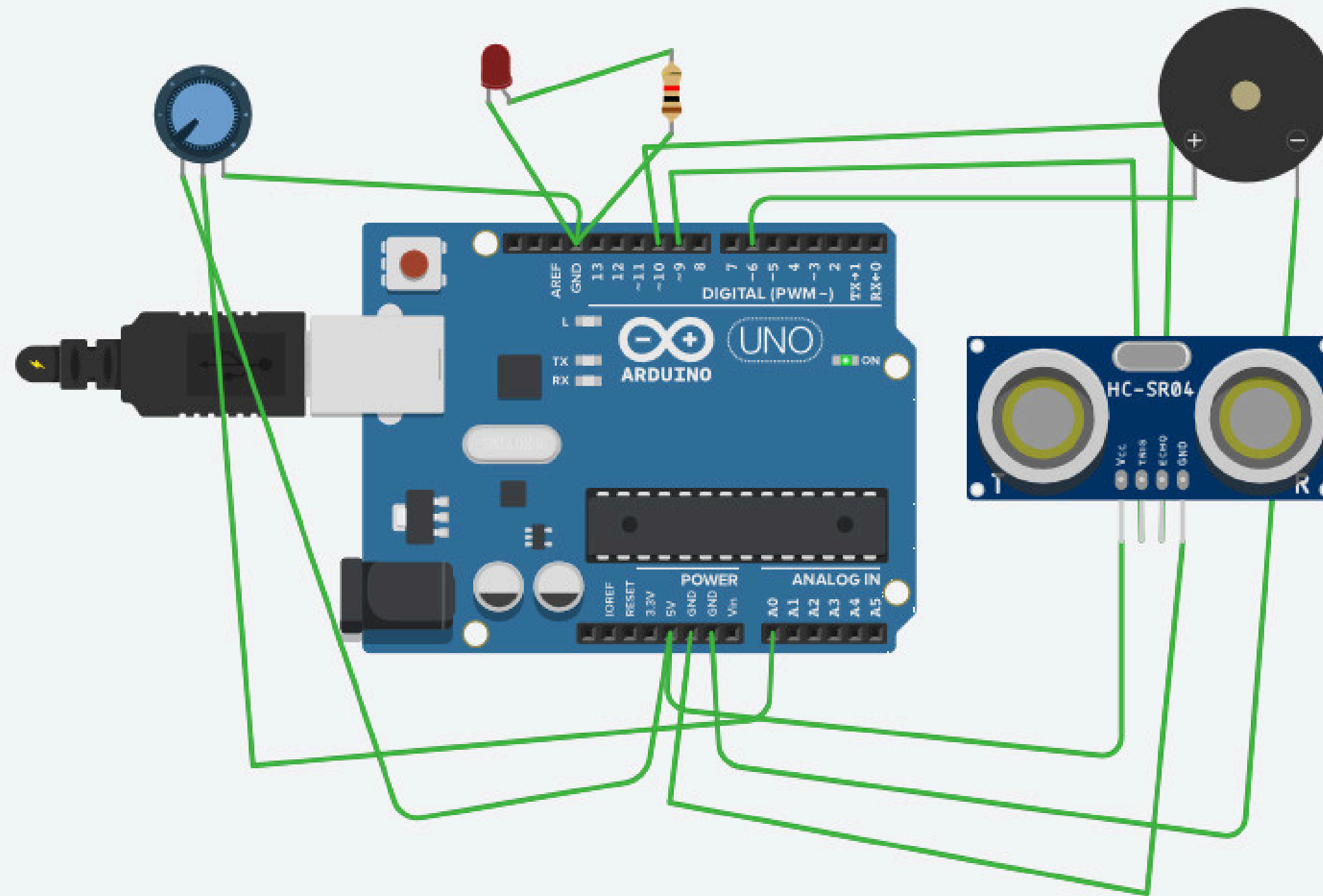
Anode (+) Pin 7 through 220Ω resistor
Cathode (–) GND

◆ Buzzer

Buzzer Pin Connect to Arduino

+ node to Pin 6
– node to GND

Circuit design



PROS AND CONS :



Pros

Gives early warnings to protect lives, property, and livestock. Shows safe places and evacuation routes. Works for local areas, not just general alerts Affordable and scalable. Solar-powered, works during power cuts.



Cons

Initial cost for sensors and setup. Needs regular maintenance. Alerts may fail if network is down. Depends on people following evacuation instructions.

MARKETING:



TARGET USERS

Municipalities,
NGOs
Disaster management
authorities
Communities in low-
lying flood-prone areas
Housing societies
schools and local farm



CHANNELS

- Partnership with local government and emergency services
- Community awareness campaigns and school programs
- Media campaigns during rainy/flood season



REVENUE MODEL

- Subscription for community-level services
- Government contracts for city or region-wide deployment
- CSR partnerships with telecom, banks, and insurance companies

CONCLUSION:

This system doesn't just monitor floods, it actively protects communities in low-lying areas by giving them early warning and actionable evacuation instructions, minimizing loss and ensuring safety.



Thank you